Emergency Water Safety and Purification Guide

This guide provides step-by-step instructions for making water safe to drink during a disaster, flood, or any emergency where tap water is unavailable or unsafe. Sourced from CDC, EPA, and WHO guidelines.

WARNING: Never drink floodwater, water with an unusual colour or smell, or water you suspect is contaminated with fuel, chemicals, or sewage. These cannot be made safe by boiling or chemical treatment alone.

SECTION 1: HOW TO KNOW IF WATER IS UNSAFE

Water is unsafe to drink after a disaster if any of the following are true:

1. Your local authority has issued a boil water notice or water advisory.
2. The water has an unusual colour (brown, green, or black).
3. The water has an unusual smell (sewage, fuel, or chemical odour).
4. The water source was flooded or may have been contaminated by floodwater.
5. You are unsure of the source. When in doubt, treat the water before drinking.

SAFE SOURCES: Sealed bottled water is always safest. Rainwater collected directly in a clean container is generally safe. Melted ice from a freezer that was sealed before the disaster is safe.

SECTION 2: METHOD 1 — BOILING (Most Reliable)

Boiling kills all bacteria, viruses, and parasites. It is the single most effective method when a heat source is available.

Step-by-step instructions:

Step 1. If the water is cloudy or has visible particles, let it settle for several minutes. Then pour the clear water from the top through a clean cloth, paper towel, or coffee filter into a clean pot. This removes sediment that can protect germs from heat.

Step 2. Place the pot on your heat source and bring the water to a full rolling boil. A rolling boil means large bubbles breaking the surface continuously — not just steam.

Step 3. Keep the water at a full rolling boil for at least 1 minute. If you are at high altitude (above 1,000 metres or 3,300 feet), boil for 3 minutes, as water boils at a lower temperature at altitude.

Step 4. Remove from heat and allow the water to cool naturally. Do not add ice to cool it unless the ice is from a safe source.

Step 5. Pour the cooled water into clean containers with lids or covers. Store away from sunlight.

Step 6. To improve the flat taste of boiled water, pour it back and forth between two clean containers several times before drinking, or add a small pinch of salt per litre.

IMPORTANT: Boiling does NOT remove chemical contamination such as fuel, pesticides, or heavy metals. If your water source may have chemical contamination, you must find a different water source entirely.

SECTION 3: METHOD 2 — HOUSEHOLD BLEACH DISINFECTION

Use this method when boiling is not possible. Use only plain unscented liquid chlorine bleach (sodium hypochlorite). Do NOT use scented bleach, colour-safe bleach, or bleach with added cleaners.

Checking your bleach:

Check the label for the percentage of sodium hypochlorite. Common concentrations are 5%, 6%, or 8.25%. This determines the number of drops needed.

Step-by-step instructions:

Step 1. Filter cloudy water first using a clean cloth or coffee filter to remove sediment. Chemical treatment works better on clear water.

Step 2. For CLEAR water: Add 8 drops of bleach per 1 litre (or per 4 cups) of water. If your bleach is 8.25% sodium hypochlorite, use 6 drops per litre.

Step 3. For CLOUDY water: Double the amount — use 16 drops per litre.

Step 4. Stir the water well to mix the bleach throughout.

Step 5. Wait 30 minutes before drinking. For very cold water (below 10 degrees Celsius), wait 60 minutes.

Step 6. After 30 minutes, smell the water. It should have a faint chlorine smell, similar to a swimming pool. If you cannot smell chlorine at all, repeat the dose and wait another 15 minutes.

Step 7. If after the second dose there is still no chlorine smell, discard this water and find a different source.

BLEACH SHELF LIFE: Bleach loses its effectiveness over time. Bleach stored for more than 1 year may be too weak to disinfect water reliably. Check the manufacture date on the bottle.

STORAGE: Treated water stored in a clean closed container is safe for up to 1 week.

SECTION 4: METHOD 3 — CHLORINE OR PURIFICATION TABLETS

Water purification tablets are available in pharmacies and chemist shops. They are compact, lightweight, and easy to use. Follow the instructions on the specific product label as concentrations vary.

General process for chlorine tablets (NaDCC, e.g. Aquatabs):

Step 1. Filter cloudy water first if possible.

Step 2. Drop the appropriate number of tablets into the water according to the label. Typically 1 tablet per 1 litre of clear water.

Step 3. Stir and wait 5 minutes for the tablet to dissolve completely.

Step 4. Wait an additional 25 minutes (total 30 minutes) before drinking.

Step 5. For cloudy or cold water, wait 60 minutes total.

IODINE TABLETS: Iodine tablets are less effective than chlorine and should NOT be used by pregnant women, people with thyroid conditions, or for more than a few weeks continuously. Chlorine tablets are preferred.

SECTION 5: WHAT PURIFICATION CANNOT DO

All of the above methods kill biological contaminants — bacteria, viruses, and most parasites. None of them remove:

- Chemical contamination: fuel, pesticides, industrial chemicals, heavy metals

- Salt: seawater or brackish water cannot be made drinkable by boiling or chemical treatment

- Radioactive particles

If you suspect chemical contamination, the only safe approach is to find a completely different water source.

SECTION 6: SIGNS OF WATERBORNE ILLNESS

If someone drinks contaminated water, symptoms may develop within hours to several days. Watch for:

- Sudden diarrhoea, especially watery or bloody

- Nausea and vomiting

- Stomach cramps and pain

- Fever and chills

- Muscle aches

DANGER — DEHYDRATION: Waterborne illness causes loss of fluids through diarrhoea and vomiting. The greatest danger, especially for children and the elderly, is dehydration. Keep giving safe water to drink even if the person is vomiting. Seek medical help urgently if the person cannot keep any fluid down or shows signs of severe dehydration: extreme thirst, dry mouth, no urination for 8 hours, sunken eyes, or confusion.